

A dataset of greenhouse gas emissions and nutrition profiles for 320 dishes from eight major cuisines of China

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Data Descriptor

Title

A dataset of greenhouse gas emissions and nutrition profiles for 320 dishes from eight major cuisines of China

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Abstract

Dietary transitions necessitate region-specific data to balance environmental sustainability and nutritional health. Recipe-level analysis offers a valuable perspective that is essential for identifying impactful ingredients and cooking practices, thereby bridging the gap between broad dietary guidelines and specific consumer behaviors. The eight major cuisines of China (including Sichuan, Cantonese, Shandong, Jiangsu, Fujian, Zhejiang, Anhui, and Hunan cuisines) exhibit remarkable diversity, incorporating a wide range of local and unique ingredients and cooking methods whose environmental impacts remain insufficiently explored. This study presents a standardized dataset detailing the carbon footprint and nutritional composition of 320 representative recipes from the eight major cuisines of China. Each recipe was normalized to a 300 g reference portion, with data collected from authoritative cookbooks and regional culinary standards. Using a life cycle assessment (LCA) framework, cradle-to-table carbon footprints were quantified across three stages: food production, post-farmgate (integrating postharvest handling, storage, processing, distribution and retailing), and cooking. Nutritional profiles were derived by matching ingredients with the Chinese Food Composition Table. The dataset provides the first comprehensive and comparable environmental and nutritional database covering all eight major Chinese cuisines and reveals significant variability in both carbon emissions and nutrient content across culinary styles and cooking methods. The dataset provides information for developing low-carbon dietary guidelines that accommodate various regional food cultures, helps the food industry optimize menus and implement carbon labeling, serves as a benchmark for environmental nutrition research, and assists consumers in making healthier and more sustainable dietary choices.

Background & Summary

Diet is a critical nexus linking the environment and human nutritional health¹. Dietary choices exert a direct impact on nutrient intake and health outcomes. Globally, dietary patterns have shifted toward a higher consumption of animal-based foods and highly processed products. This trend has been linked to an increased prevalence of chronic conditions, such as obesity, cardiovascular diseases, and diabetes, as well as growing environmental pressures from food systems^{2,3}. Between 1997 and 2022, China's consumption of animal-based foods such as pork, poultry, and beef and mutton continued to rise, while the intake of plant-based foods including: vegetables, fruits, dairy products, and legumes remained insufficient. The per capita carbon footprint of food consumption increased by 31-44%⁴. Given the dual imperative to reduce environmental impact and maintain nutritional balance in dietary optimization, numerous organizations and national bodies have developed corresponding dietary guidelines⁵. An increasing number of countries have begun to incorporate cultural diversity into their dietary guidelines. Brazil's guidelines center on foods and meals, use meal examples derived from local dietary data, and systematically advocate for cultural practices such as cooking skills⁶. As tangible expressions of culinary culture, coupled with the growing availability of data on their environmental and nutritional characteristics, recipes can serve as a critical foundation for improving such guidelines. Environmental and nutritional data for recipes are essential for developing more culturally adaptive dietary guidelines and providing consumers with practical guidance on food choices.

The eight major cuisines of China refer to China's eight major regional culinary traditions, each with its own distinctive local flavors, including Sichuan (Chuan), Cantonese (Yue), Shandong (Lu), Jiangsu (Su), Fujian (Min), Zhejiang (Zhe), Anhui (Hui), and Hunan (Xiang) cuisines⁷. Each region has developed its own culinary culture with distinct cooking methods and flavors based on its unique climate, geography, history, availability of produce, and dietary customs⁸, as shown in Table 1. Different cuisines exhibit unique systems in ingredient selection⁹, cooking techniques, and flavor philosophies¹⁰, leading to considerably different environmental impacts and health implications. Yue cuisine emphasizes the original taste and freshness of ingredients, predominantly uses seafood and poultry, and favors slow-cooked soups, braised dishes, and steaming. While slow simmering enhances flavor, the longer cooking time results in higher energy consumption, leading to a larger proportion of carbon emissions from the cooking stage compared to other cuisines¹¹, and higher levels of purines and fat¹². Long-term consumption could be associated with elevated uric acid and lipid levels, potentially increasing the risk of cardiovascular diseases, diabetes, and gout^{13,14}. Chuan cuisine frequently features pork, beef, and bean products, often prepared through high-temperature stir-frying, which requires a short cooking time, resulting in a lower proportion of carbon emissions in the cooking stage¹¹. Chuan cuisine utilizes large amounts of oil and complex seasonings, resulting in its signature numbing, spicy, and aromatic profile¹⁵, its emissions related to seasonings are higher than those of Cantonese cuisine¹¹. Although stir-frying may relatively reduce cooking energy consumption, the high-salt, high-oil, and high-spice dietary pattern is often linked to hypertension and gastrointestinal burdens¹⁶. It is essential to analyze the complete recipe, encompassing ingredient composition and cooking, to assess both carbon footprint and nutritional profiles. A gap remains in the availability of nutritional and environmental data for regional cuisines, which hinders a precise understanding of regional-scale culinary cultures and their environmental impacts in the Chinese diet.

Table 1 Characteristics of the eight major cuisines of China.

Cuisine	Region	Key ingredients	Example dishes
Chuan Cuisine	Sichuan	Pork, beef, bean products, chili, Sichuan	Mapo Tofu,

(Sichuan)		pepper	Kung Pao Chicken
Yue Cuisine (Cantonese)	Guangdong	Seafood, poultry, vegetables	Steamed Fish, Dim Sum
Lu Cuisine (Shandong)	Shandong	Seafood, grains, vegetables	Sweet and Sour Carp, Moo Shu Pork
Su Cuisine (Jiangsu)	Jiangsu	Freshwater fish, hairy crabs, bamboo shoots, ham	Braised Pork Ball, Beggar's Chicken
Min Cuisine (Fujian)	Fujian	Seafood, mushrooms, red yeast rice, satay sauce	Buddha Jumps Over the Wall
Zhe Cuisine (Zhejiang)	Zhejiang	Fish and shrimp, bamboo shoots, Longjing tea leaves	Dongpo Pork, Longjing Shrimp
Xiang Cuisine (Hunan)	Hunan	Pork, pickled chilies, smoked meats	Steamed Fish Head with Chili, Stir-fry Pork with Chili Peppers
Hui Cuisine (Anhui)	Anhui	Fermented fish, hairy tofu, black pig belly pork	Stinky Mandarin Fish, Mao Tofu

Existing research has focused on the cradle-to-table environmental impacts and health implications of diets at the recipe level in countries such as China¹¹, Japan¹⁷, the UK, the USA and other countries¹⁸. However, such studies are often confined to macro-level national assessments, overlooking the heterogeneity of culinary traditions^{18,19} or are limited to specific Chinese cuisines (e.g., Chuan and Yue cuisines)²⁰. Currently, there is still a lack of health and environmental analysis covering all regional cuisines, yet these data are crucial for formulating culturally appropriate dietary guidelines. Given the considerable heterogeneity of Chinese culinary practices and unique cooking techniques (i.e., stir-frying), accurately estimating carbon emissions during the cooking stage is complex. Existing studies on quantifying the environmental impacts of the cooking stage have primarily relied on measured energy consumption for specific dishes¹¹ and simplified energy consumption models based on standardized cooking equipment (e.g., estimating energy consumption as power multiplied by time) that do not distinguish between cooking methods²¹. The standardized assumptions overlook the diversity of Chinese cooking techniques, such as stir-frying, stewing, steaming, and deep-frying, which differ in terms of heat transfer and energy efficiency. To address this gap, methods for quantifying energy consumption of different cooking techniques should be adopted to more accurately capture the carbon footprint variations arising from diverse Chinese cooking techniques.

This study aims to establish a standardized dataset of carbon footprints and nutritional compositions for representative dishes from China's eight traditional cuisines. By focusing on the micro-level recipes, the study reveals how differences in ingredient selection and cooking techniques across the eight cuisines influence carbon emissions and nutritional density. This study fills a gap in sustainability assessments tailored to China's regional food cultures and refines the description of carbon emissions characteristics associated with Chinese cooking techniques. The dataset is widely applicable and available for use by various stakeholders. Policymakers can use the dataset to develop low-carbon dietary guidelines that reflect regional culinary cultures. The food

industry can use it for menu optimization and carbon labeling. Researchers can use it as a foundation for studies on the environmental impacts of Chinese diets and sustainable dietary patterns.

Methods

Overview of the framework

The framework for constructing a standardized dataset of carbon footprint and nutrition estimation for the eight major cuisines of China follows these systematic steps (Figure 1). First, mainstream recipes from the eight major cuisines of China were collected, and the ingredient names, weights, cooking methods, and cooking times of the recipes were standardized. Second, the life cycle assessment (LCA) method was applied to quantify the cradle-to-table carbon footprint of the recipes, encompassing the following stages: food production, postharvest handling and storage, processing, distribution, retailing, and cooking stages. The ingredients in the recipes were cross-referenced against the China Food Composition Table to calculate their nutritional content. This study constructed a standardized dataset detailing the nutritional composition and environmental footprint of dishes from eight major cuisines of China.

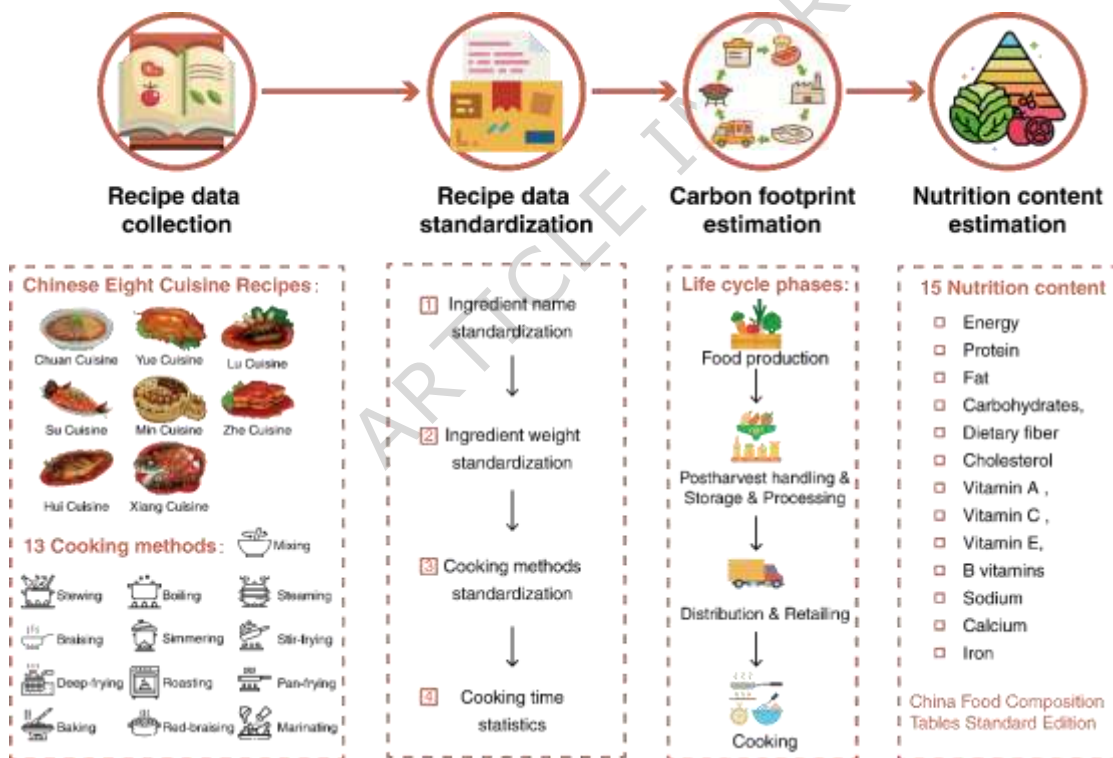


Figure 1 Framework of carbon footprint and nutrition estimation for eight major cuisines of China.

Recipe selection

The recipe data for the eight major cuisines of China, including Sichuan (Chuan), Guangdong (Yue), Shandong (Lu), Jiangsu (Su), Fujian (Min), Zhejiang (Zhe), Hunan (Xiang), and Anhui (Hui) cuisines, were sourced from authoritative cookbooks and regional culinary standards containing with recipe information and cooking methods, including Chinese Eight Regional Cuisines²², Chinese Famous Recipes²³, Technical Specifications for Classic Sichuan Cuisine²⁴, and the Shandong Cuisine Standard System²⁵. To ensure the selection of representative

and popular dishes, this study identified mainstream dishes from each of the eight Chinese cuisines based on the 2023 White Paper on Chinese Cuisine²⁶. The Baidu Index, an indicator reflecting public interest based on online search frequency, was used to determine the most popular dishes from the eight cuisines. Based on Baidu Index data for December 2023, the top 40 classic recipes with the highest search volumes were selected from each cuisine, compiling a dataset of 320 recipes that comprehensively covers the eight great traditional cuisines in China. This multi-step selection process, combining authoritative cookbooks, industry white papers and public search data, ensures that the collected recipes reasonably reflect mainstream consumer preferences across all eight cuisines.

Recipes Standardization

Information on each recipe within the dataset, including the names and weights of all ingredients, along with the cooking method and time, was systematically recorded. These data points for components, quantities, and techniques were compiled from standard reference cookbooks and relevant regional specifications. The names of all ingredients appearing in the recipes were standardized against the food names and their corresponding food categories in the China Food Composition Table. The food categories comprised non-ruminant meats, ruminant meats, aquatic products, vegetables & fruits, oils, water & soup, seasonings, grains, and beans & nuts. To quantify the carbon footprint of dishes per serving per person in China, the ingredient weights in each recipe were standardized to a reference portion of 300 g. This value was consistent with previous studies¹¹ and was cross-checked against the Chinese Dietary Guidelines (2022)²⁷ and the Report on Chinese Residents' Nutrition and Chronic Diseases (2020). According to these references, the recommended daily intake of non-cereal food was 576–1,011 g, with lunch and dinner each accounting for approximately 35% of daily intake, yielding an estimated per-meal dish weight range of 202–354 g; the 300 g value fell within the per-meal dish weight range.

The collected recipes span 13 cooking methods: stewing, boiling, steaming, braising, simmering, stir-frying, deep-frying, roasting, pan-frying, baking, red-braising, marinating, and mixing. When a recipe involves multiple cooking methods, the cooking technique was identified as the core step that determines the final texture and flavor of the dish, as described in the authoritative cookbooks and regional culinary standards. This identification was verified by manual cross-checking. Cooking times that were not explicitly specified in published standards were calculated as the average of the cooking times from identically named recipes collected from the online recipe platforms “Meishijie” (<https://www.Meishi.cc>)²⁸ and “Meishichina” (<https://www.meishichina.com/>)²⁹ in March 2022. As the Meishijie website has been redesigned, the URL no longer provides access to the recipe data. Detailed recipe information is now available on the mobile platform after authentication.

Carbon footprint assessment

This study conducted a life cycle assessment (LCA) in accordance with ISO 14040/14044 guidelines to quantify the cradle-to-table carbon footprint of 320 recipes from the eight major cuisines of China. This study aimed to quantify the carbon footprint of dishes per serving per person for the eight major cuisines of China and reveal the differences among the eight cuisines. The system boundary was quantified across three stages: food production, post-farmgate (integrating postharvest handling, storage, processing, distribution and retailing), and cooking, as shown in Figure 2. The carbon footprint was defined as the total greenhouse gas (GHG) emissions, quantified in carbon dioxide equivalents (CO₂e) by converting carbon dioxide (CO₂), methane (CH₄), and nitrous oxide (N₂O) using the 100-year Global Warming Potential (GWP) from the Intergovernmental Panel on Climate Change (IPCC). Fluorinated gases were not considered in this assessment. The functional unit was defined as

300g, representing the weight of a per-serving dish for one person per meal. The consumption point was defined as urban households in China, as the cooking energy mix used was based on Chinese urban data. Ingredients constituting less than 5% of the total recipe weight were excluded from the carbon footprint calculations due to their negligible contribution and the lack of reliable emission factors. For the production stage, all ingredients were assumed to be produced within China, and emissions from foreign production stages and imported ingredients were not considered. Food losses and waste occurring at life cycle stages were excluded from the assessment, as China's food waste mainly occurs during post-cooking consumptions which fall outside the system boundaries in this study³⁰.

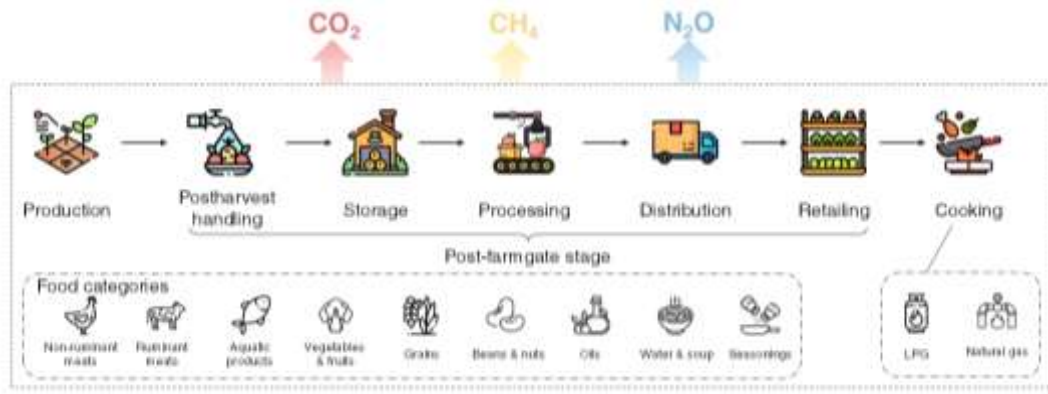


Figure 2 System boundary of eight major cuisines recipes.

This study applied the life cycle assessment (LCA) to quantify the cradle-to-table carbon footprint of the 320 recipes from eight major cuisines of China. The total carbon footprint for each recipe i was calculated by summing the contributions from all stages, as expressed in the following equation:

$$CF_i = CF_{pi} + CF_{ti} + CF_{ci} \quad (1)$$

where, CF_i (gCO₂e) represents the average carbon footprint of recipe i , and CF_{pi} , CF_{ti} , CF_{ci} denote the carbon footprint from the production, post-farmgate (including postharvest handling and storage, processing, distribution, and retailing), and cooking stages.

(1) Food production stage

For the food production stage, the carbon footprint CF_{pi} (gCO₂e) of recipe i was determined using Equation 2.

$$CF_{pi} = \sum_j W_{ij} \times F_{pj} \quad (2)$$

where W_{ij} is the weight(g) of ingredient j , and F_{pj} (gCO₂e/g) is the specific production stage emission factor. The F_{pj} values are drawn upon established sources: the database consolidated by Petersson, et al.³¹ and the Chinese Life Cycle Greenhouse Gas Emission Coefficients for Products³². A concordance table linking standard ingredient names to food categories has been stored in the Figshare repository under the name “Greenhouse gas emission factors”. For the few ingredients without available emission factor data, the average greenhouse gas factor of ingredients within their respective food category was used as a substitute.

(2) Post-farmgate stages

The post-farmgate stages were divided into two aggregated stages: one covering postharvest handling, storage, and processing, and the other covering distribution and retailing. The carbon footprint CF_{ti} (gCO₂e) of recipe i from the post-farmgate stages were determined using Equation 3.

$$CF_{ti} = \sum_j W_{ij} \times (F_{aj} + F_{bj}) \quad (3)$$

where W_{ij} is the weight (g) of ingredient j , F_{aj} (gCO₂e/g) represents the GHG emission factor for postharvest handling, storage, and processing stages, and F_{bj} (gCO₂e/g) represents the GHG emission factor for distribution and retailing stages. These GHG emission factors are averages calculated for a food category and for ingredient j . The values for F_{aj} and F_{bj} were sourced from Xue, et al.³³, who integrated 868 life cycle assessment studies from the BCFN Double Pyramid Database (2016). These studies were compiled from research conducted in China between 2008 and 2016 and represent the multi-year national average GHG emission factors for the Chinese food system.

(3) Cooking stage

To quantify the carbon footprint during the cooking stage, this study adopted the methodological framework and all parameters as described in Inaba, et al.³⁴ and Long, et al.¹⁷. Energy consumption was estimated for all 13 cooking techniques, and corresponding GHG emission factors for energy sources were used to calculate the final footprint. Given that natural gas and LPG are the primary cooking fuels in urban households in China³⁵, energy use of ten methods, including boiling, braising, stewing, simmering, steaming, stir-frying, baking, pan-frying, red-braising, and deep-frying, was modeled based on household stoves powered by fossil fuels (natural gas and LPG), with the relative contributions of natural gas and LPG set at 82.5% and 17.5%³⁶.

$$CF_{ci} = \frac{E_i \times F_n \times b_n}{h_n} + \frac{E_i \times F_l \times b_l}{h_l} \quad (4)$$

where CF_{ci} (gCO₂e) denotes the cooking stage carbon footprint for recipe i . E_i (J) is the total energy consumed during cooking, with technique-specific calculations provided in Equations 6–14. The terms F_n (2.8 kgCO₂e/Nm³) and F_l (5.11 gCO₂e/g) correspond to the emission factors for natural gas and LPG, sourced from reference³². Their respective household usage proportions, b_n (82.5%) and b_l (17.5%), were from the Statistics³⁶, while the calorific values h_n (35,608.5 kJ/m³) and h_l (50,242 kJ/kg) were drawn from the national standard³⁷.

For dishes prepared by roasting, energy consumption was modeled as originating from electric ovens. The cooking emissions were derived from:

$$CF_{ci} = \frac{E_i \times F_e}{h_e} \quad (5)$$

where F_e is the Chinese 2023 grid electricity emission factor (530 gCO₂e/kWh)³² and h_e is its calorific value (3,602.005 kJ/kWh)³⁷. Marinating and mixing methods involve no thermal energy input; thus, their cooking stage carbon emissions were designated as zero.

The subsequent section details the energy consumption models for each cooking technique.

1) Boiling, Braising, Stewing, and Simmering

The total energy required for these cooking methods encompasses three distinct parts: the energy needed to bring water to a boil, to sustain the boiling state, and to raise the temperature of the food ingredients. The corresponding calculation formulas are provided below:

$$E_b = E_{bw} + E_{baw} + E_{bh} \quad (6)$$

$$E_{bw} = \frac{m_w \times c_w \times \Delta k_b}{\eta h} \quad (7)$$

$$E_{baw} = \frac{v \times S_p \times H_w \times \Delta T}{\eta b} \quad (8)$$

$$E_{bh} = \frac{\sum_k Q_k \times c_k \times \Delta k_b}{\eta b} \quad (9)$$

where E_b (J) denotes the total energy consumption, comprising three components: E_{bw} (energy to boil water), E_{baw} (energy to maintain boiling), and E_{bh} (energy to heat ingredients). Key parameters include: the mass of water m_w (g), set at 5/3 of the ingredient weight based on culinary practice; the specific heat capacity of water c_w (4.18 J/(g • K)); and the temperature rise Δk_b (80 K from 20° C to 100° C). The thermal efficiencies for heating and boiling are ηh (0.37) and ηb (0.42), respectively. The evaporation rate v is 0.06 g/(min • cm²) over a pot base area S_p of 314 cm², with H_w (2250 J/g) as the latent heat of vaporization. ΔT represents the post-boiling heating time, assumed to be two-thirds of the total duration. For the ingredients, Q_k (g) is mass and c_k (J/(g • K)) is specific heat capacity.

2) Steaming

For steaming, energy consumption was calculated using Equations 6-9. The model was based on a standard household steamer with a depth of 15 cm and an inner diameter of 30 cm. The mass of water m_w was parameterized as 60% of the total pot volume.

3) Stir-Frying, Baking, Pan-Frying, and Red-braising

The energy consumption associated with these dry-heat methods was determined by the following equation:

$$E_s = \frac{\sum_k Q_k \times c_k \times \Delta k_f}{\eta h \times \alpha} \quad (10)$$

where E_s (J) represents the total energy consumption, Δk_f denotes the increase in ingredient temperature (set at 100 K), and α is a correction factor with a value of 0.75, which models the heat loss from food in open cookware¹⁷.

4) Deep-frying

For deep-frying, the total energy consumption E_f (J) comprises two components: the energy required to heat

the oil, termed E_{fo} (J), and the energy needed to heat the ingredients, denoted E_{fh} (J):

$$E_f = E_{fo} + E_{fh} \quad (11)$$

$$E_{fo} = \frac{m_o \times c_o \times \Delta k_o}{\eta h} \quad (12)$$

$$E_{fh} = \frac{\sum_k Q_k \times c_k \times \Delta k_d}{\eta h} \quad (13)$$

where m_o and v are the mass (g) and specific heat capacity (2.0 J/(g·K)) of the cooking oil. The term Δk_o (160 K) corresponds to the oil's heating from 20°C to 180°C, while Δk_d (130 K) models the resultant temperature rise of the food being fried.

5) Roasting

For baking in an electric oven (Panasonic NB-HM3810 model), the energy consumption E_r was modeled using the following calculation:

$$E_r = P_m \times \Delta T_m \quad (14)$$

where P_m denotes the rated power of the oven (1800 W); ΔT_m represents the total baking time in seconds.

Nutritional Composition Estimation.

The nutritional profile of each recipe was assessed across fifteen components: energy; macronutrients including carbohydrates, protein, and fat; vitamins comprising Vitamin A, vitamin C, vitamin E, and B vitamins; minerals such as calcium and magnesium; as well as dietary fiber and cholesterol. We further evaluated each recipe's contribution to the recommended daily intake of these nutrients. The calculation is based directly on raw ingredient composition, without factoring in cooking-related nutrient losses. The content N_{ic} of nutrient c in a given recipe was computed as follows:

$$N_{ic} = \sum_j \frac{W_{ij} \times P_j \times O_{jc}}{100} \quad (15)$$

where P_j refers to the edible portion coefficient (in percentage) for ingredient j , and O_{jc} indicates the amount of nutrient c per 100 g of ingredient j . Both parameters were obtained from the China Food Composition Table³⁸.

Data Records

The comprehensive dataset of recipe nutrition and carbon footprint is publicly available at Figshare³⁹ as a detailed spreadsheet. It is organized into three sheets. The “Recipe” sheet includes the English and Chinese dish names, cooking methods, cooking time, ingredients, carbon footprint (covering the production, post-farmgate, and cooking stages), and nutritional metrics such as energy, protein, fat, carbohydrates, dietary fiber, cholesterol, vitamins A, C, E, B₁, and B₂, as well as sodium, calcium, and iron. The “Greenhouse gas emission

factors” sheet lists ingredient names and their corresponding food types and is used for data matching during life-cycle carbon footprint accounting. The “Data_source” sheet provides the sources for all data types used in this study.

Technical Validation

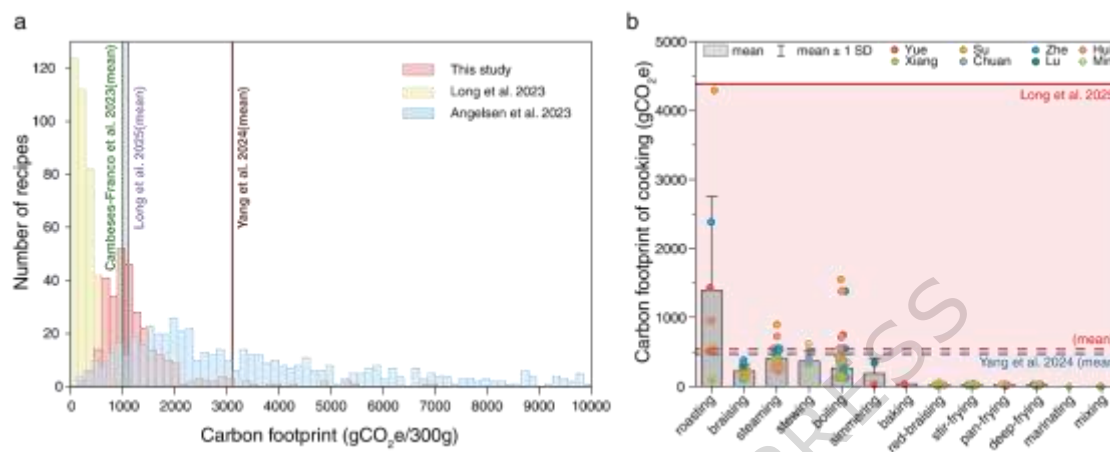


Figure 3 Comparative distributions of carbon footprints across recipes in this study and other literature sources. The dashed line in Figure 3b represents the mean carbon footprint from existing literature, and the corresponding colors indicate the range of results.

Statistical analysis. The carbon footprints of recipes from the eight major cuisines of China in this study range primarily from 267 to 6930 g CO₂e per 300 g, with an average carbon footprint across the 320 recipes of 1337 g CO₂e /300 g. To validate the data from this study, we aggregated existing carbon footprint data for recipes, standardizing all recipe weights to 300 g for comparison^{17,19,20,40}, as shown in Figure 3a. Long, et al.²⁰ quantified the average carbon footprint of 1000 Sichuan and Cantonese cuisine recipes as 1107 g CO₂e / 300 g, which is close to the average carbon footprint of Sichuan and Cantonese cuisines (1232 g CO₂e /300 g) in this study. The distribution of recipes in this study falls entirely within the range reported in previous research.

As depicted in Figure 3b, the cooking stage carbon footprints of the eight major cuisines in this study mainly range from 0 to 4767 g CO₂e, with an average cooking stage carbon footprint of 696 g CO₂e. The zero values correspond to marinating or mixing recipes, which require no heating and thus produce no cooking stage emissions. In the study by Yang, et al.⁴⁰ the average cooking stage carbon footprint for the Chinese cuisines was 473 g CO₂e. Long, et al.²⁰ quantified the cooking stage carbon footprints of Cantonese and Sichuan cuisine recipes within a range of 0–4388 g CO₂e, with an average of 543 g CO₂e. The distribution of cooking stage carbon footprints in this study is consistent with the ranges reported in existing research.

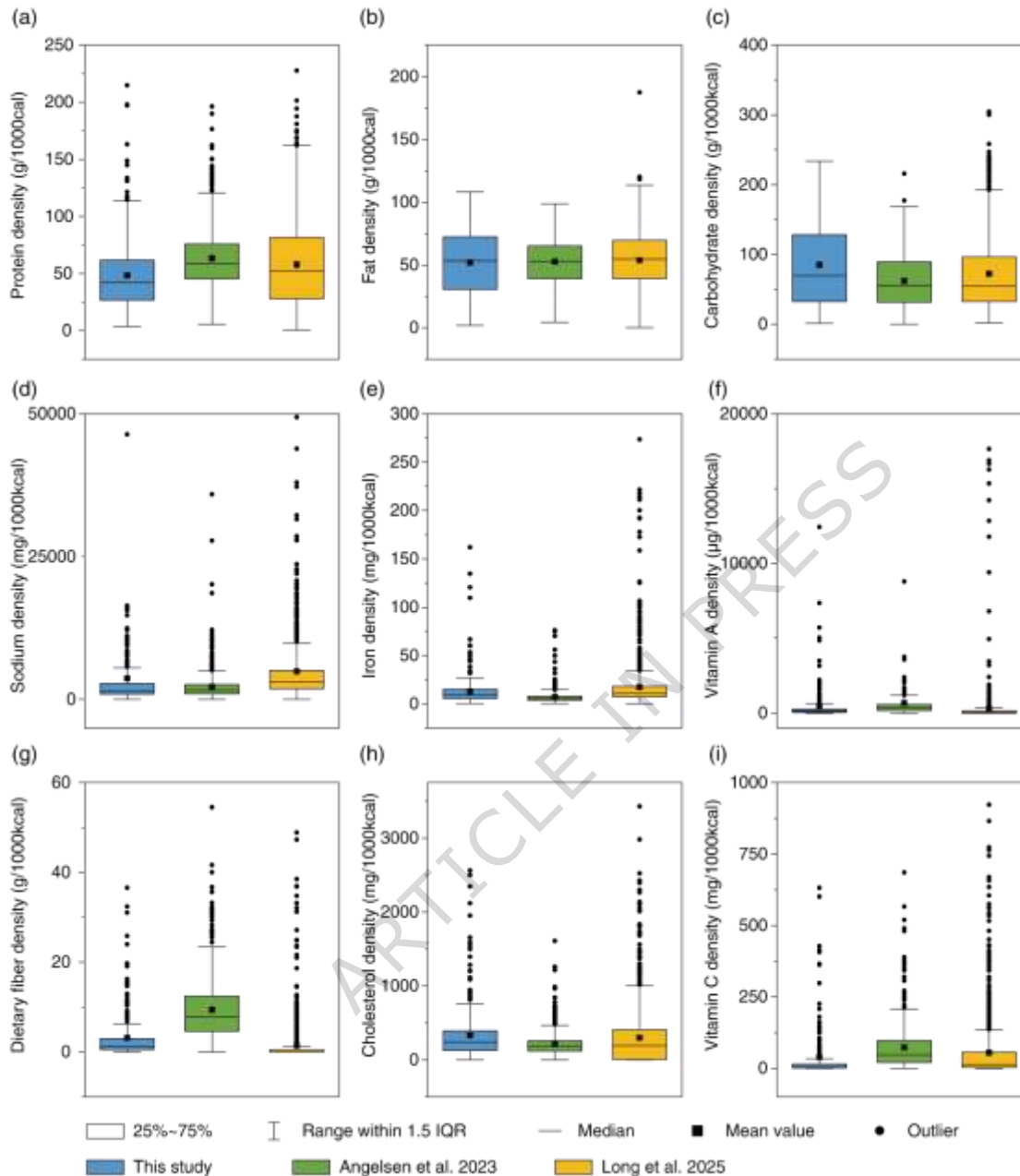


Figure 4 Comparative distributions of nutrient density across recipes in this study and other literature sources

The validation of recipe nutrition was performed by comparing nutritional density per unit energy. Based on previous studies, the nutrient contents of the recipes were uniformly normalized to 1000 kcal, thereby enabling comparison independent of portion size^{20,41}. As shown in Figure 4, previous studies (e.g., Angelsen et al., 2023, covering recipes from Norway, the UK and the USA; and Long et al., 2023, covering Sichuan and Cantonese cuisines in China) had reported that the nutritional densities were 0.5 - 227.6 g/1000 kcal for protein, 0.2 - 187.3 g/1000 kcal for fat, and 2.1 - 487.3 g/1000 kcal for carbohydrates. For the eight major Chinese cuisines, the average nutritional density ranges for protein, fat, and carbohydrates were 3.6–214.8 g, 1.9–108.1 g, and 1.3–233.7 g per 1000 kcal, respectively. This demonstrates that the nutritional composition of recipes from the eight major cuisines generally falls within the nutrient ranges reported in multiple studies.

Uncertainties and Limitations. This study has some limitations that may lead to uncertainty. For example, minor ingredients such as spices and condiments were excluded from the scope of carbon accounting due to the lack of reliable emission factors. This may result in an underestimation of the carbon footprint, particularly for seasoning-intensive cuisines such as Chuan cuisine. Future work should incorporate emission factors for these ingredients when they become available to improve the accuracy of the quantification. The cooking energy mix was modeled based on natural gas and LPG, which represent the dominant urban household energy structure in China. Electricity and other alternatives were not included, as they currently have a limited share in urban cooking. The results do not reflect potential shifts under electrification policies. Future work should explore alternative energy scenarios, such as fully electric cooking or varied fuel proportions, to assess their impacts on GHG emissions during the cooking process. GHG emissions under urban and rural energy structures could be compared to assess the sensitivity of the results to this assumption. To facilitate the estimation of cooking-stage emissions, this study adopted a simplified approach by assigning a primary cooking technique to each recipe. It is acknowledged that this approach may not fully capture the complexity of the dish flavors. Future work should aim to incorporate uncertainty quantification, refine the treatment of minor ingredients, account for nutrient changes during cooking, and model dishes prepared with multiple cooking techniques more accurately. The results are applicable to the Chinese context and not transferable to other countries, since the GHG emission factors and cooking energy mix are both based on Chinese data. The methodological framework can be adapted to other regions by substituting local emission factors or recipe data.

Data Availability

The dataset is available at Figshare (<https://doi.org/10.6084/m9.figshare.32721765.v1>).

Code availability

No custom code was used in the study.

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Author contributions

Y.Z. led the project and Y.Z. and D.G. designed the study. A.G. and Z.C. collected the data and conducted the calculations. Y.Z. and A.G. validated the dataset. A.G. conducted the analyses and visualization and drafted the manuscript. All authors contributed to the revision of the manuscript.

Competing interests

The authors declare no competing interests.

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